

A closed-form implied-volatility correction at a cubic local-volatility knot pinned at the money

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Abstract

Practitioners push a parametric local volatility σ_{loc} through the leading-order Berestycki–Busca–Florent (BBF) inverse-harmonic-mean map and close the gap with the Henry–Labordère $\sigma_1 T$ correction (PHL1) or the Gatheral–Hsu–Laurence–Ouyang–Wang $\sigma_2 T^2$ refinement (GHLOW2). These work while σ_{loc} is smooth. A piecewise-cubic σ_{loc} is only C^2 at its knots — the cubic coefficient $\gamma = \sigma_{\text{loc}}'''$ jumps — and that single broken derivative produces a localised implied-volatility error that none of the smooth- σ_{loc} asymptotics capture. We give the closed-form PHL1 correction for the special case of a knot pinned exactly at-the-money. At first order in the coefficient jump it is a one-dimensional Brownian-bridge integral $\Phi_{BB}(x, 0)$ with peak $3\sqrt{2\pi}/128$. The at-the-money restriction is what makes the result clean: the general case requires an unbounded-polynomial truncation to stay finite, which the $w = 0$ specialisation removes. We validate against the Dupire PDE on parameters calibrated from real SPXW surfaces.

1 Setup

Work in log-moneyness $k = \log(K/F)$, forward measure ($F = 1$, $r = 0$), single maturity. Local volatility is one cubic, in annualised % units,

$$\sigma_{\text{loc}}(k) = \sigma + \beta k + \alpha k^2 + \gamma k^3. \quad (1)$$

A knot at the money adds a one-sided jump δ in the cubic coefficient,

$$\sigma_{\text{loc}}(k) = \sigma + \beta k + \alpha k^2 + \gamma k^3 + \delta k^3 H(k), \quad (2)$$

with H the Heaviside step. Because k^3 and its first two derivatives vanish at $k = 0$, (2) is C^2 at the knot; only σ_{loc}''' jumps by δ . This is precisely the structure a piecewise-cubic calibrator produces, with the knot moved to ATM.

We use the Black $T = 1$ convention; scale = $\sqrt{252/n} \cdot 100$ maps total vol to annualised % for $n = \max(1, \text{DTE})$ business days, and $\sigma_{\text{total}} = \sigma/\text{scale}$ is the unperturbed ATM total vol.

The asymptotic maps. The leading-order BBF map [Berestycki et al., 2002] is the inverse harmonic mean

$$\text{iv}_{\text{hm}}(k) = k / \int_0^k \frac{dy}{\sigma_{\text{loc}}(y)}, \quad \text{iv}_{\text{hm}}(0) = \sigma_{\text{loc}}(0), \quad (3)$$

whose use as a parametric local-vol-to-implied-vol representation, and its structure, are studied in detail by De Marco [De Marco, 2021]. The next-to-leading heat-kernel expansion of Henry–Labordère [Henry–Labordère, 2008] adds a $\sigma_1 \cdot T$ term, giving $\sigma_{\text{IV}} \approx \text{iv}_{\text{hm}} + \sigma_1$. We use the explicit

per-strike closed form of that correction as stated by Gatheral, Hsu, Laurence, Ouyang and Wang [Gatheral et al., 2012, Thm. 2.4], and keep the conventional label **PHL1** for it:

$$\sigma_1(k) = \frac{\text{iv}_{\text{hm}}(k)^3}{2k^2} \log \frac{\sqrt{\sigma_{\text{loc}}(0) \sigma_{\text{loc}}(k)}}{\text{iv}_{\text{hm}}(k)}. \quad (4)$$

(Henry-Labordère’s own local-to-implied first-order result is the same correction in the equivalent Hagan–Woodward midpoint form.) The GHLOW2 refinement adds a further $\sigma_2 T^2$ term from the same source [Gatheral et al., 2012]. All three are smooth wherever σ_{loc} is smooth and inherit a kink at a knot, since (4) sees σ_{loc}''' .

2 The at-the-money knot correction

The implied-vol response to (2) expands in the jump δ about the knotless case. At first order, in dimensionless strike $x = k/\sigma_{\text{total}}$,

$$\delta\sigma_{\text{IV}}(k) = \delta \cdot \sigma_{\text{total}}^3 \cdot \Phi_{BB}(x, 0) + O(\delta^2). \quad (5)$$

Very simple derivation. First order in δ , the source is $v(y) = 2\sigma \delta y^3 H(y)$. Inserting it once into the Duhamel expansion of the Dupire call price against the constant- σ Black heat kernel, the Black strike-gamma identity cancels the explicit σ , and a Gaussian completion turns the spatial integral into an expectation under a Brownian bridge pinned at $(0, 0)$ and (k, T) : writing $\lambda = t/T$, the bridge has $y \sim \mathcal{N}(\lambda k, \sigma_{\text{total}}^2 \lambda(1 - \lambda))$. The remaining y -integral against $y^3 H(y)$ is a truncated third moment of a normal,

$$f(\eta) = (\eta^3 + 3\eta) \Phi(\eta) + (\eta^2 + 2) \phi(\eta), \quad (6)$$

with Φ, ϕ the standard normal cdf/pdf. The $\sqrt{\lambda}$ from the Gaussian completion cancels the $1/\sqrt{t}$ of the strike-gamma, the outer time integral collapses to a clean bridge-time integral, and dividing by the Black vega gives (5) with

$$\Phi_{BB}(x, 0) = \int_0^1 (\lambda(1 - \lambda))^{3/2} f(\eta(\lambda; x)) d\lambda, \quad \eta(\lambda; x) = x \sqrt{\frac{\lambda}{1 - \lambda}}. \quad (7)$$

At the knot centre $\eta \equiv 0$, $f(0) = 2/\sqrt{2\pi}$, and the integral is a Beta function:

$$\Phi_{BB}(0, 0) = \frac{2}{\sqrt{2\pi}} B\left(\frac{5}{2}, \frac{5}{2}\right) = \frac{3\sqrt{2\pi}}{128} \approx 0.05875. \quad (8)$$

Why at-the-money. The general knot at offset $w = k_{\text{knot}}/\sigma_{\text{total}}$ keeps a factor $\eta = [\lambda x - (1 - \lambda)w]/\sqrt{\lambda(1 - \lambda)}$; the subtraction below then leaves a polynomial that grows without bound and needs an ad-hoc truncation to stay finite. Setting $w = 0$ removes the $(1 - \lambda)w$ term entirely and the correction is bounded with no hack — the reason this note restricts to the ATM knot.

Avoiding double counting. PHL1 already moves under (2) through its own iv_{hm} and σ_1 . Taylor-expanding (4) in δ at $w = 0$ gives, for $x > 0$, iv_{hm} varies as $x^3/4$ and σ_1 as $x/4$ (both zero

for $x \leq 0$). Subtracting these so they are not counted twice yields the *directed* kernel, which decays on both sides:

$$\Phi_{BB}^{\text{dir}}(x, 0) = \begin{cases} \Phi_{BB}(x, 0) - \frac{x^3}{4} - \frac{x}{4}, & x > 0, \\ \Phi_{BB}(x, 0), & x \leq 0. \end{cases} \quad (9)$$

The corrected method, with PHL1 evaluated on the perturbed surface so the cancellation is exact to first order, is

$$\sigma_{\text{IV}}^{\text{PHL1+corr}}(k) = \sigma_{\text{IV}}^{\text{PHL1}}(k) + \delta \cdot \sigma_{\text{total}}^3 \cdot \Phi_{BB}^{\text{dir}}(k/\sigma_{\text{total}}, 0). \quad (10)$$

The bridge integral (7) is evaluated with a 32-point Gauss–Legendre rule.

3 Results

Parameters come from real SPXW surfaces calibrated with a 20-parameter piecewise-cubic local-vol model; we keep the centre cubic and (for the knot case) the first-knot coefficient jump moved to ATM. The Dupire forward PDE (Rannacher start then Crank–Nicolson) is the ground truth.

Figure 1: a smooth monotone-skew smile (SPXW 2025-03-10 DTE 1, no knot). BBF0 is ~ 67 bps from the PDE across the band; PHL1 and GHLOW2 reduce that to a few bps — each order helps.

Figure 2: a concave (half-circle) converted smile (DTE 3). The approximation remains good where vol is positive; the deep wings, where the option’s time value underflows and implied vol is not recoverable, are trimmed rather than faked.

Figure 3: the same DTE 1 cubic with a fake ATM knot (δ the real first-knot jump, moved to $k = 0$). PHL1 is visibly biased near the knot; PHL1 plus Φ_{BB}^{dir} tracks the PDE.

4 Reproducibility

All numbers and figures are produced by the open TypeScript core in the companion repository, cross-checked against an independent Python reference to 10^{-6} (closed forms) and 10^{-9} (the kernel). The figures are deterministic: regenerating them and diffing reproduces this paper exactly.

Credits. Market data for the calibrated example smiles: ThetaData.

References

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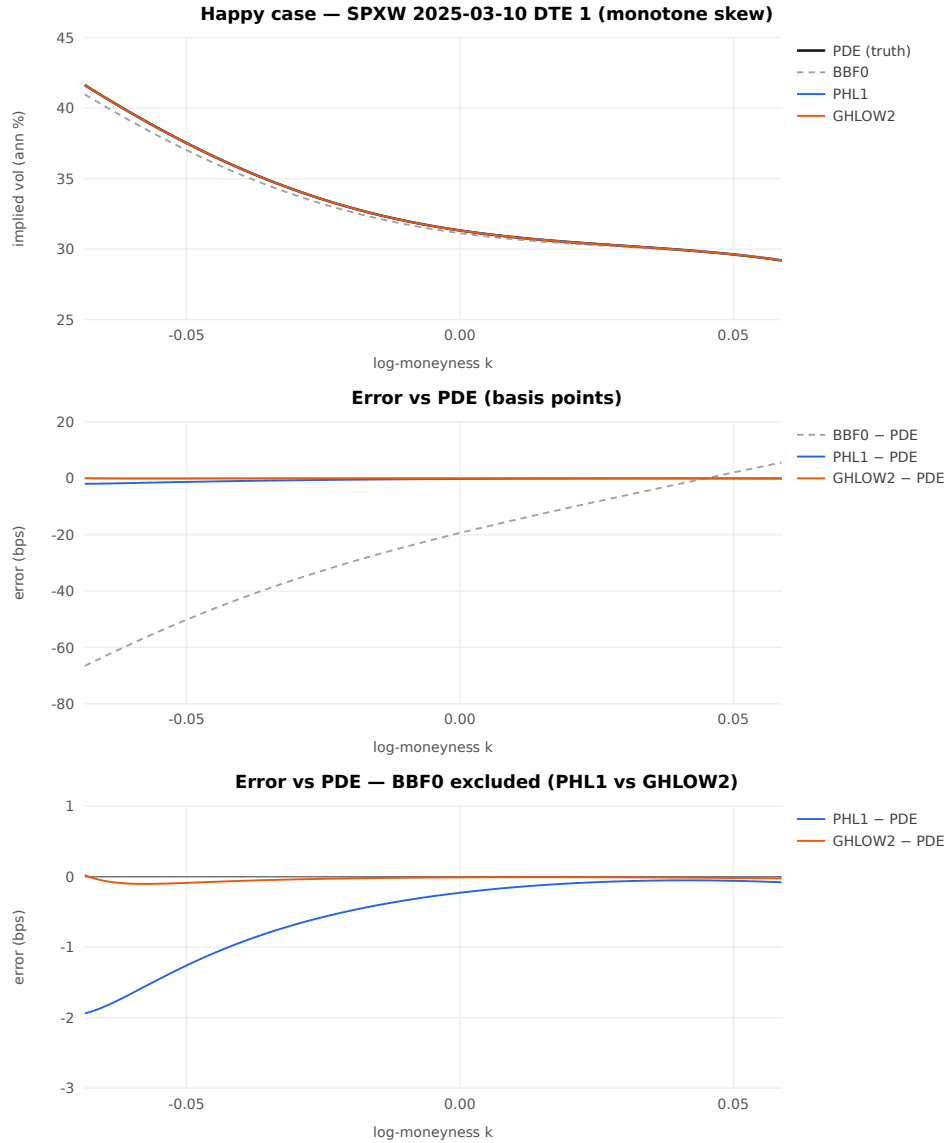


Figure 1: Happy case: smile (top), error vs PDE in bps with BBF0 (middle), and the same error with BBF0 excluded (bottom) so the PHL1-vs-GHLOW2 comparison is legible.

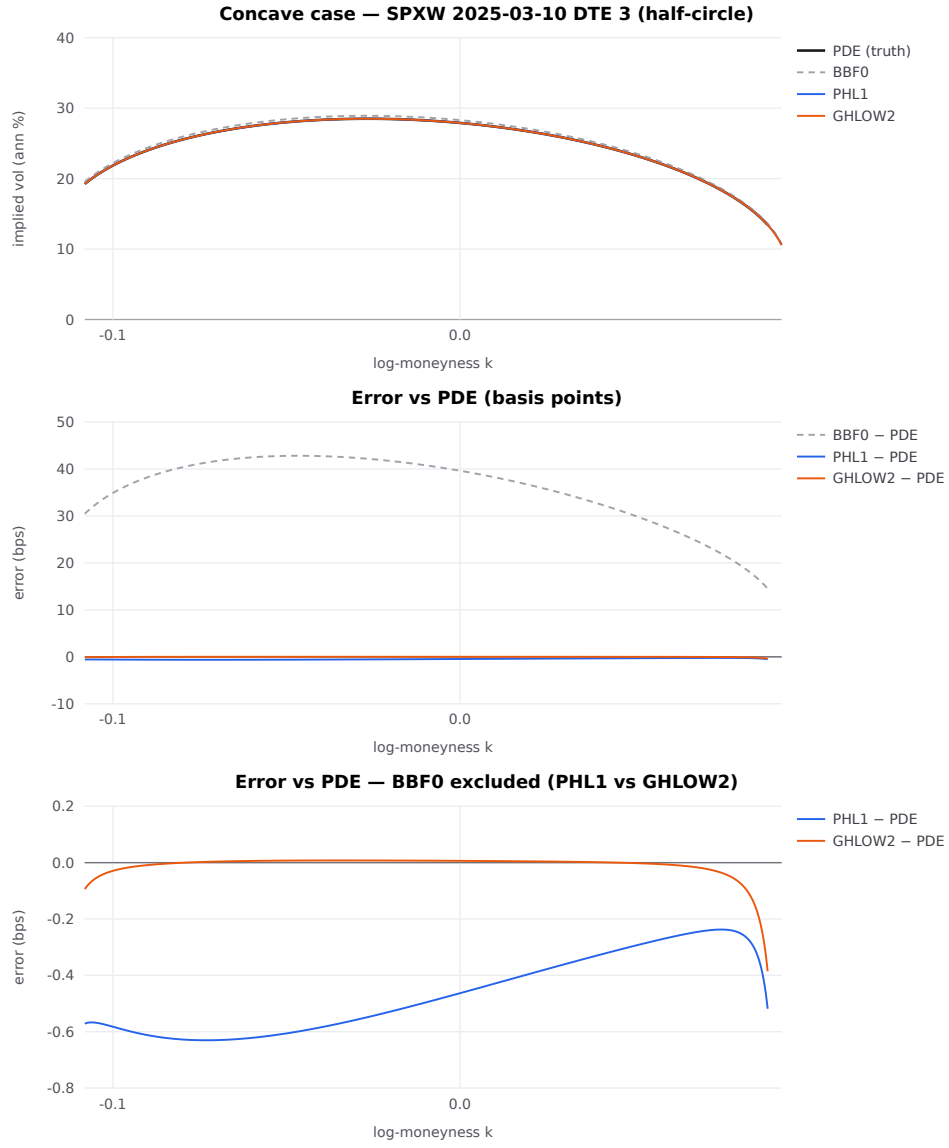


Figure 2: Concave case: the converted smile is a half-circle. Panels as in Figure 1 (smile; error with BBF0; error without BBF0).

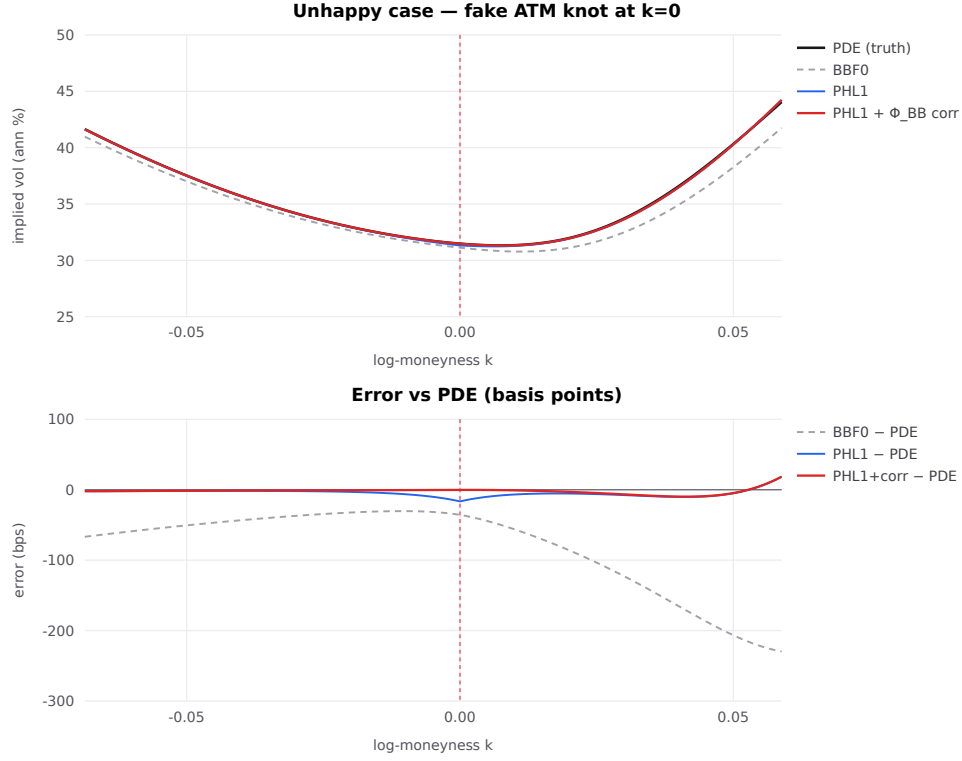


Figure 3: Unhappy case: a fake at-the-money knot. PHL1+correction recovers the PDE that PHL1 alone misses near $k = 0$.

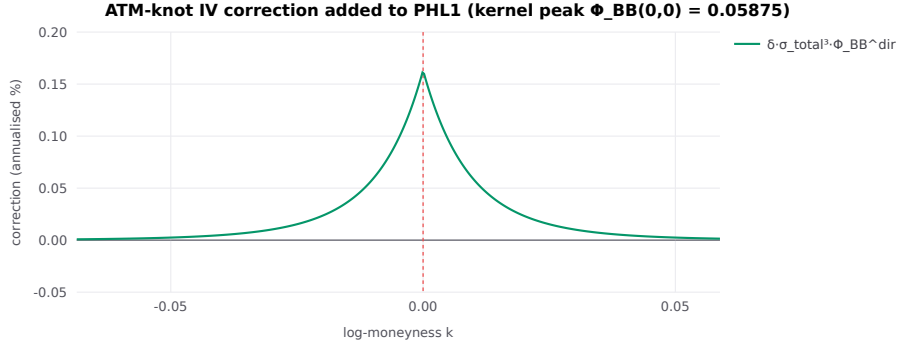


Figure 4: The applied ATM-knot correction $\delta \sigma_{\text{total}}^3 \Phi_{BB}^{\text{dir}}(k/\sigma_{\text{total}}, 0)$ in annualised % for the knot case (exactly the gap added to PHL1 in Figure 3). It is the dimensionless directed kernel — peak $\Phi_{BB}(0, 0) = 3\sqrt{2\pi}/128$, decaying on both sides after the PHL1 subtraction — scaled by $\delta \sigma_{\text{total}}^3$.